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REVIEW ARTICLE

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A comprehensive exploration of artificial intelligence competence frameworks for educators: A critical review

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Abstract

Recent literature underscores the need for teachers to develop AI competencies with a recognition of the current lack of well-defined competence frameworks. This critical review investigates teachers' Artificial Intelligence (AI) competence frameworks (AI CFTs), analysing their strengths, weaknesses and practical applications for researchers, educators and policymakers. It identifies five distinct types of AI CFTs within Competence Construct Claims (Child, S., & Shaw, S. 2023). A conceptual approach to validating competence frameworks. *Research Matters: A Cambridge University Press & Assessment publication*, *35*, 27–40.), each addressing the complexities of AI in its early stages. Notably, frameworks derived from empirical data offer detailed descriptions of competencies, while those based on conceptual models provide broader overviews. Highlighting the need for further empirical research, this review helps identify and understand existing approaches to teacher AI competence development and paves the way for integrating AI CFTs into teacher education, ultimately enhancing educators' preparedness to harness AI in their teaching practices.

KEYWORDS
AI competence frameworks for teachers, AI competencies for educators, AI in education, competence models

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1 | INTRODUCTION

Artificial Intelligence (AI) has brought transformative changes in education, offering innovative tools and approaches that can enhance teaching and learning (Chakravorti, 2022; Gill et al., 2023; Grubaugh & Levitt, 2023). AI in education promises to support students with special needs, reduce school dropout, personalise learning, support decision-making, generate feedback, assess new skills, etc. (Bentley et al., 2018; Brundage, 2019; Vincent-Lancrin & van der Vlies, 2020). Additionally, AI in education (AIED) brings a fresh set of competencies that educators should cultivate. The rapid advancement of AI technologies requires an understanding of AI capabilities and limitations. This entails more than just basic digital literacy; it encompasses data literacy, algorithm literacy, computational thinking and design thinking. Notably, the ethical use of AI in educational settings becomes a paramount concern, making AI and data ethics a crucial aspect of teacher competence (Berendt et al., 2020; U.S. Department of Education, Office of Educational Technology, 2023; European Commission, 2022).

Even as schools increasingly embrace AI tools, teacher preparation is not prioritised and is experiencing delays. A recent UNESCO global survey, which included over 450 schools and universities, revealed that fewer than 10% of them have established institutional policies or formal guidance regarding the use of generative AI applications (UNESCO survey, 2023).

The rise of AI in education has provided food for discussion on teachers' AI competencies in recent literature.2020 Trujillo-Cabezas (2020) proposed AI digital competencies for educators, including the utilisation of applications, adept information management, content creation for learning and facilitating student connections through technology. Teachers require both knowledge and tools to evaluate the appropriateness of AI-supported activities in achieving their educational objectives and transform their way of teaching (Flogie & Krabonja, 2023). Liu and Li (2022) defined teacher human capital for AI in education through three facets: expertise and proficiency in AI in education, emotional disposition and attitudes relevant to AI in education, and ethical considerations for using AI in education. Xu (2020) underscores educators developing AI competency for transformative roles, while Markauskaite et al. (2022) support integrating digital technologies and gaining hands-on AI experience.

Despite the growing interest in AI competencies, the existing literature highlights a significant gap in well-defined frameworks for educators to acquire competencies essential for using AI to the benefit of their learners. Ng et al. (2021a) emphasise the limited exploration within current studies of enhancing teachers' AI digital competency through teacher education programmes. Polak et al. (2022) conclude that, while examining various AI frameworks, the development of AI competencies involves a broad spectrum of skills, knowledge and attitudes. While some are specific to AI, others are adapted versions of digital competencies or essential competencies irrespective of technology use (such as problem-solving and contextual consideration). Ng et al. (2022) reflect on three educational theories; Bloom's taxonomy to view cognition levels of AI literacy education, the TPACK model to understand how educators should learn to equip their students with AI literacy (Ng et al., 2021b), and the P21's Framework to identify the four types of digital competencies in the 21st century. AI digital competencies, as outlined by the P21's Framework for 21st Century Learning (National Research Council, 2012), recognise AI literacy as a vital skill within a comprehensive framework. This perspective emphasises its role in fostering non-technical, critical and complex literacies. Regardless of these early attempts to analyse the AI CFTs, the emergence of new AI frameworks highlights challenges in understanding their purpose and utilising and adapting them for professional development programmes.

Currently, there are several critical issues related to AI CFTs that have been not addressed by previous study of Ng et al. (2022) (see, Table 1). First, there might be a scattered landscape of various AI CFTs, making it difficult for educators, researchers and policymakers to find and compare them efficiently. Second, different AI CFTs might exhibit varying levels of quality and effectiveness in preparing educators to utilise AI. Third, there might be a need for a deeper understanding of how AI CFTs are designed and how they can practically be incorporated into teacher education programmes. Finally, educators might not feel adequately prepared to leverage AI in their teaching practices due to a limited understanding of relevant frameworks.

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TABLE 1 Comparison of studies on AI CFTs.

Studies discussing several AI competence/ literacy frameworks for teachers	Problems addressed	Contribution	Advantages	Disadvantages
Ng et al. (2022) Inclusion of AI literacy in educational theories (AI included in Bloom's taxonomy, TPACK, P21's Framework)	• Lack of Established Frameworks • Challenges for Educators	• Formulated theoretical basis to investigate how AI literacy should be implemented • Conceptual Framework mapping different cognitive levels • Integration Strategy to bring AI literacy into schools and broaden AI Literacy to 21st-century skill	• Holistic understanding of different learning levels and practical integration methods for AI literacy • Advocating for AI literacy for everyone	• Limited scope: Other conceptual frameworks are not analysed • Evolving field: ongoing revisions to the framework with the evolving nature of AI technologies
Our study analysis of available AI CFTs in recent literature Higuera (2019), Kim and Kwon (2023), Lorenz and Romeike (2023), Ethical Guidelines on the Use of AI, 2022 Olari and Romeike (2021), UNESCO (2023), Holstein et al. (2020), Molenaar (2022), Carolus et al. (2023), Su and Yang (2023), Sattelmaier and Pawlowski (2023), Ng et al. (2023)	• Lack of understanding of AI CFTs • Difficulty in choosing and adapting frameworks • Limited understanding in preparing educators for using AI	• Compilation and categorisation of AI CFTs • Enhanced understanding and practical application of frameworks • Identifying gaps in existing frameworks and future directions • Facilitating discussion and collaboration by bringing together various AI CFTs	• Facilitation of broader understanding, informed selection and adaptation of AI CFTs • Exploration of the characteristics of various frameworks	• Limited scope: Other conceptual frameworks may emerge • Subjectivity in analysis

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Therefore, this critical review addresses problematic issues related to the effective integration of AI in education. The review represents the initial effort to compile teachers' AI competence frameworks, analyse their strengths and weaknesses, and categorise them to enhance their practicality for researchers, educators and policymakers. Exploring various AI CFTs may help to understand their design and characteristics, and facilitate their integration into teacher education enhancing the preparedness of educators in harnessing AI potential within their teaching practices.

The paper explores the new AI CFTs that emerged in the recent educational literature, analyses them within the Competence Construct Claim (Child & Shaw, 2023), and classifies groups of AI CFTs into five distinct types: (1) Existing competence model-oriented; (2) Competence areas/domains-oriented; (3) process-driven (process competency model); (4) AI systems-driven (specific work/job model); and (5) Competence level-driven (multidimensional). Each type of CFT reflects a specific underlying approach to modelling teachers' competencies and how these models could effectively be applied to prepare teachers for AI in education. The paper highlights several pertinent questions and challenges educators face when they want to embrace AI in their practices; it considers the importance of theoretical models in the successful integration of AI in education and the interplay between subject-specific knowledge and AI proficiency.

1.1 | Research objective of the review

- To explore and classify AI CFTs into distinct types or approaches within Competence Construct Claims category (Child & Shaw, 2023). This objective seeks to provide a structured framework for understanding the definitions, labels and rationale of different AI CFTs.

The research questions

The following questions guided the literature review:

RQ 1: Which AI CFTs are available in recent educational literature?

RQ 2: How can AI CFTs be classified within Competence Construct Claims? What are underlying approaches, focus areas and characteristics of these AI CFTs?

To address the research objectives and research questions, we searched for the AI CFTs in scientific literature online in Scopus and Web of Science with the key words 'Artificial Intelligence', 'Competence framework' or 'Competence models', 'for teachers' or 'for educators'. Next, the papers were screened to check their suitability for the topic and analysed according to the Construct Claim (Child & Shaw, 2023).

By addressing these research objectives and research questions, this study aimed to provide a classified overview of AI competence frameworks that are designed to equip teachers with essential skills. Furthermore, it highlights the existing challenges in these frameworks, thereby contributing to the ongoing dialogue on how educators can best prepare for AI in education.

2 | EDUCATORS HAVE CHALLENGES IN INTEGRATION AI IN EDUCATION

Educators are one of the key actors in the integration of AI in education in the present revolutionary era of Industry 4.0 (Felix, 2020). This era is marked by 'connecting everything to everything everywhere' – a vast interconnected network of devices, autonomous vehicles, smartphones and smart systems designed for sensing, predicting, decision-making and interaction with the physical world (Rothman & Gulli, 2022, p. 12). Teachers, being pivotal in preparing the next generation for an AI-integrated future, play a crucial role in fostering willingness and

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readiness for AI implementation in schools, ultimately equipping students for a collaborative human-AI future (Ayanwale et al., 2022). The success of AI education depends on teacher readiness, as the teaching and learning process requires their central involvement (Trotsko et al., 2019). According to ethical guidelines developed by (European Commission, 2022), it is important for educators to understand the basic mechanisms and limitations of AI systems and how AI systems can be used to support teaching and learning in a safe and ethical way.

AI has the potential to support teachers and improve the educational experience for students. It encompasses various areas such as AI instruction, intelligent systems, adaptive learning and gamification (Hwang et al., 2022; Zawacki-Richter et al., 2019; Zhai et al., 2021). Bryant et al. (2020) suggested that 20 to 40 percent of current teacher time spent on activities could be automated using existing technology to save teacher time. For example, AI real-time analytics tools can help teachers monitor and assist students using AI-based tutoring software for self-paced learning (Aleven et al., 2022). Additionally, AI systems can assist teachers by automatically tracking and recognising students' motions and behaviours, providing valuable insights for personalised guidance (Felix, 2020). AI technologies can also tailor education to students' needs and automate routine tasks, allowing teachers more time for creativity (Chikobava & Romeike, 2021). AI systems can do automated, repetitive task and design, that empower learning experiences, creativity, thinking, solutions of problems, collaboration and they will augment teachers' roles. Supporters of AI in education stress that AI will perform repetitive tasks and pattern recognition, which are constantly done by teachers (Arora, 2021).

The integration of AI technologies into educational contexts requires not only teachers' digital skills but AI competencies as well (Sostero & Tolan, 2022; Williamson & Eynon, 2020). Educators' AI competencies include a range of skills that empower teachers to critically evaluate AI technologies, effectively engage with AI, and employ AI tools in diverse settings. According to Edwards et al. (2018), educators need the competence to choose and create instructions aligned with machine learning, track students' advancement and offer emotional and cognitive assistance through robotics powered by artificial intelligence.

However, equipping teachers with AI competencies seems to lag behind the optimism around the use of AI in daily educational practice. Many teachers feel overwhelmed by the pressure to integrate AI into their teaching without fully understanding its workings and impact (especially with recent rapid developments in the field of Large Language Models). AI concepts (Computer vision, machine learning, deep learning, natural language processing, etc.) tend to be abstract and complex and are hard to understand for teachers (Kim & Kwon, 2023; Long & Magerko, 2020); there are scarce AI professional development opportunities for teachers, very few institutional policies or formal guidance on the use of generative AI applications and inadequate guidance on the necessary competencies for educators in academic literature and existing educational policies (Ayanwale et al., 2022; UNESCO, 2023).

To empower educators with AI competencies, it is essential to delineate a targeted set of knowledge domains and skills, systematically arranged within a competency framework or model. The goal of this AI competency framework for educators is to articulate fundamental proficiencies (Knowledge, skills and abilities) essential for teachers. This competency framework can serve as a tool for program developers, and learning coordinators involved in the program development process, from developing early ideas to implementation (Blok et al., 2021).

For successful integration of AI in education, teachers, as primary stakeholders in education and in continuous interaction with their learners, should be engaged in an early phase. As practical guidance on how to equip teachers with AI competencies is lacking, a first step may be to develop a framework or model that aligns AI competencies necessary to use AI to the benefit of learners.

3 | AI COMPETENCE FRAMEWORKS FOR EDUCATORS ARE CLASSIFIED

In this section, we will define competencies, competence frameworks and the approach to framework design. Then we will discuss AI CFTs classified according to five categories: (1) Existing competence model-oriented; (2) Competence areas/domains-oriented; (3) process-driven (process competency model); (4) AI systems-driven (specific work/job model) and (5) a competence level-driven (multidimensional).

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The educational approach to defining competencies focuses on skill development, achieving standards and obtaining credentials. A competence can be narrowly defined as an action, behaviour, demonstrated outcome or minimum standard, with varying levels of mastery articulated through different statements (Bourke et al., 1975 cited in Markus et al., 2005). The OECD defines competencies more broadly and beyond mere knowledge and skills: '[Competence] involves the ability to proficiently tackle complex challenges by utilising and deploying psychosocial resources, encompassing both skills and attitudes, within a particular context' (OECD,2005 , p. 4). In this paper, we refer to OECD' definition of 'competence'.

Competencies in education, like in industrial enterprises, are often grouped under a competence framework or model, which outlines essential skills and behaviours needed for effective performance in specific roles or educational contexts. Competency modelling is the activity of determining specific competencies that are characteristic of high performance and success in a given job. Organisations adopting a competency approach create or utilise a competency model, which at minimum consists of a simple list or catalogue specifying desirable competencies (Markus et al., 2005).

Organisational competency models come in different types and structures, depending on the institutions' objectives, values and belief system (Chouhan & Srivastava, 2014; Krajcovicova et al., 2012). Frameworks include descriptions of competencies organised into tiers, making it easier for educators and professionals to understand and apply them to their work. Various competency models have been created with diverse categorisation of competencies: personal, social/interpersonal, action-related and domain-related, or grouping competencies into meta, domain, method and social competencies. Additionally, there are competency models specifically designed for leadership and management, as well as for particular tasks or job profiles (Prifti et al., 2017). Additionally, the models can be designed with two approaches: top-down (by experts – competency modelling), bottom-up (suggested skills by people working in the field – job analysis). Competency modelling emphasises future roles aligned with a strategic plan and defines optimal performance through worker attributes. In contrast, job analysis primarily describes current work activities and establishes typical performance without addressing alignment issues within the organisation (Stevens, 2013).

A competence framework makes a claim about its interrelated purpose, creating a position that requires validation. Experts examine claims by questioning them and gathering evidence (Child & Shaw, 2023). From an analysis of the competence framework literature, Child and Shaw (2023) identified four categories of claims in the competence framework: Construct Claims, utility claims, audience claims and impact claims. In this review, we focus on Competence Construct Claims of the competence frameworks, which are related to the definitions of competencies 'how elements are defined and labeled; how they combine to constitute overall competence; the rational for defining and organizing things in this way rather than some other way' (p. 31).

We searched for AI CFTs, studied them and noted the similarities among frameworks. Next, we categorised them in a classification scheme within a Competence Construct Claims to provide a structured approach for understanding the definitions, labels and rationale of different AI CFTs: (1) Existing competence model-oriented; (2) Competence areas/domains-oriented; (3) Process-driven (process competency model); (4) AI systems-driven (specific work/job model); and (5) Competence level-driven (multidimensional). The framework categories N2, 3, 4 and 5 were found in the scientific literature (Klieme et al., 2003; Prifti et al., 2017; Rothwell & Lindholm, 1999). Below, we will examine each of these types of AI CFT. First, we will explain the claim and the definition of the framework, then, we will provide an example of the framework that falls into this category; and finally, we will summarise the characteristics and strengths of each model.

3.1 | Integrating AI competencies in existing CFTs

The development of an AI competence framework finds strong support in established theoretical models and competence frameworks for teachers, with notable applicability observed in the Technological Pedagogical Content

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Knowledge model (TPACK) and existing competence frameworks for teachers, such as the Digital Competence Framework for Educators (Mishra & Koehler, 2006; Punie, 2017), or the Information Communication Technology Competence Framework for Teachers (ICT CFT) (UNESCO, 2018). These competence models have already been implemented and validated in practice (Hooker et al., 2011; Schmidt et al., 2009). Thus, they offer structured approaches to incorporating competencies into education. The following questions help us examine these frameworks: How can new AI competencies be included in the earlier frameworks? What links can be identified between existing competencies and the new ones?

In this category, we included Higuera's (2019) framework, who proposed five pillars or competencies that should be added to teacher training in AI:

- *Uncertainty and randomness* – The acknowledgement of data inconsistency and the non-causal nature of AI-generated forecasts in education. It underscores that AI predictions are estimations rather than absolute truths. Embracing uncertainty implies teaching without assuming omniscience and accepting the reality that mistakes can occur.

- *Coding and computational thinking* – The justification for including these lies in their prevalence in curricula in many countries. Coding involves utilising programming language libraries, emphasising the importance of coding skills. The argument is made that proficient AI users must comprehend algorithms to discern when it is necessary not to rely blindly on machine learning decisions.

- *Data awareness* – Data and computing are at the centre of computing. Most effort will concern data collection, visualisation, manipulation and analysis.

- *Critical thinking* – Achieving true technological literacy involves a fusion of comprehending digital concepts and utilising algorithms to form informed opinions. The goal is to teach not just the use of technology but also to impart the ability to understand its impacts, assess risks and cultivate a historical perspective on media.

- *Post AI humanism* – Impact on non-technological attitudes, reconsider some ground truth: Truth, experience, creativity and intelligence.

Higuera (2019) combined these five pillars with six aspects of the UNESCO ICT competence framework for teachers (ICT-CFT, UNESCO, 2018) by assessing the effectiveness of the ICT-CFT as a framework for training teachers in AI and measures that ICT-CFT would benefit from the impact of AI. The relationship between pillars and aspects is described in three ways: mutual impact, the impact of the pillar on the aspect and the influence of an aspect on the pillar (see Table 2). The primary goal of the ICT CFT is to equip educators with hands-on expertise in effectively using and developing educational materials through ICT, which serves as the basis to integrate understanding of computers, algorithms and data within AI.

Another important model, the Digital Competence Framework for Educators (2017), undergoes changes with regard to AI. The EU Commission has added a new pivotal competence within this European framework: the ethical utilisation of AI and data. For ethically employing AI systems, educators are required to understand essential aspects such as *human agency and oversight*, *transparency*, *diversity*, *non-discrimination*, *fairness*, *societal and environmental well-being*, *privacy*, *data governance*, *technical robustness*, *safety and accountability*. These elements are included within the framework of digital competencies in the context of the European Framework for the Digital Competence of Educators (Ethical Guidelines, 2021).

Kim and Kwon (2023) began with the creation of a prototype of an AI competency framework for elementary school teachers through a literature review. This prototype was refined with empirical data on teachers' experiences, specifically focused on AI teaching practices. The final result included a list of 22 AI competencies

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TABLE 2 Integration of AI pillars in UNESCO ICT-CFT adapted from Higuera (2019).

Mutual impact of aspects and pillars	Impact of pillars on the aspects	Influence of aspects on pillars
A4. Application of digital skills Randomness Data awareness	Coding and Computer thinking A1. Understand ICT in Education	A1. Understand ICT in Education Post AI Humanism
	Computer thinking and coding A3. Pedagogy	A2. Curriculum and assessment Post AI Humanism
	Randomness A4. Application of digital skills	A2. Curriculum and assessment Critical thinking for AI
	Coding and Computer thinking A5. Organisation and administration	A3. Pedagogy Data awareness
		A6. Teacher professional learning Post AI Humanism

TABLE 3 AI competencies included in TPACK framework adapted from Kim and Kwon (2023).

Content knowledge (CK)
• Understanding the basic foundations of CS and STEM • Understanding the basic concepts of AI • Understanding AI ethics • Understanding the copyright while using open educational resources (OER) • Understanding knowledge related to the AI workforce
Pedagogical knowledge (PK)
• Managing the classroom in an online and in-line class • Facilitating project-based learning or inquiry-based learning • Providing timely feedback and encouraging peer-review • Overcoming the fear of failure and trial and error during the class
Technological knowledge (TK)
• Searching for AI learning tools or materials and using them in the class • Using AI learning tools or learning materials in the class • Openness to AI and new technologies
Pedagogical Content knowledge (PCK)
• Designing the AI curriculum with core subjects • Organising AJ tools and learning resources according to students' developmental stage • Motivating students with daily topics or problems that they are interested in
Technological Content knowledge (TCK)
• Literacy in code and data • Literacy of AI technologies in daily life • Training AI models using literacy in a dataset • Being sensitive to social impacts or problems from AI
Technological Pedagogical knowledge (TPK)
• Using AI tools for the assessment • Organising students' AI artefacts and learning outcomes using online platforms and digitisation techniques for portfolios • Demonstrating AI-related practices using learning tools and materials.

tailored for teachers in elementary school settings, categorised as sub-competencies within the Technological Pedagogical Content Knowledge (TPACK) framework (Mishra & Koehler, 2006). The AI competencies for teachers included competencies categorised as CK, PK, TK, PCK, TCK and TPK (see Table 3). Teachers' Content Knowledge (CK) in AI and computer science principles is meant to alleviate anxiety and enhance confidence in teaching AI

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concepts, and the integration of Pedagogical Knowledge (PK) and Technological Knowledge (TK) in teacher professional development programmes is likely to improve instructional design and the incorporation of AI education into classrooms.

Lorenz and Romeike (2023) have systematically characterised teaching-related AI competence areas by adapting the Digital Pedagogical and Content Knowledge (DPACK) model. They justify the incorporation of AI-related digital competencies within this framework, replacing the traditional 'T' for Technology with 'D' for Digital and situating AI competencies within the Technology/Digital domain. The framework introduces three key competencies: AI-K (Knowledge), AI-PK (Pedagogical Knowledge), AI-CK (Content Knowledge) and AI-PCK (AI Related Pedagogical Content Knowledge). AI-K involves the ability to recognise, understand, reflect and shape AI phenomena from technological, socio-cultural, and user-oriented perspectives. AI-PK focuses on recognising and reflecting on the potentials, limitations and risks of AI in teaching-learning processes, facilitating the design of contemporary educational settings. AI-CK pertains to recognising and reflecting on the implications of increased AI use in one's discipline, affecting the scientific field, professional domain and subject matter. AI-PCK involves recognising and reflecting on the subject-specific influence, potentials and limitations of AI in teaching and learning processes and learners, guiding the design of modern educational settings. Based on the DPACK model, the framework provides a structured description of AI education requirements.

While existing theoretical models such as TPACK and ICT CFT serve as valuable foundations for new competence frameworks, it is essential to acknowledge that they may have certain rigidities that can limit the integration of innovative elements unless they are modified. Nevertheless, they provide a solid starting point for researchers, educators and policymakers to develop comprehensive AI competence frameworks for educators.

3.2 | Competence areas/domains-oriented (competency structured models)

Another common approach to developing a competence framework is to delineate the framework's areas by formulating key questions or defining specific knowledge domains: What are the new knowledge areas? What are the essential new skills or attitudes that educators need? These areas encompass mandatory skills for teachers and various facets of education, ranging from classroom management to the ethical utilisation of AI, and they serve as the primary focal points for policymakers and educators.

Olari and Romeike (2021) examined the relationship between AI and data literacy competence areas in existing educational frameworks and proposed a preliminary approach that utilises the data lifecycle to reflect on data literacy competencies relevant to AI. They found that the available AI competence frameworks developed for teacher education and schools do not include data literacy. To include data literacy, they proposed a novel approach comprising eight primary competencies: (1) Extracting information from datasets and processing with AI tools (Acquisition stage); (2) Conducting effective data cleaning and mitigating biases in datasets (Cleansing stage); (3) Integrating learning processes and data management systems into practical tasks grounded in AI concepts (Implementation stage); (4) Enhancing data through the application of AI techniques (Optimisation stage); (5) Utilising AI algorithms to visualise data (Visualisation stage); (6) Generating new insights from data through AI-based analyses (Analysis stage); (7) Explaining results, including identifying errors and addressing them through questioning (Evaluation stage); and (8) Employing AI-driven procedures to enhance data archiving, deletion, or sharing processes (Sharing, erasing and archiving stages).

Another significant document related to AI CFT is an ongoing draft document by UNESCO. The organisation develops education from pre-school to higher education and beyond, strengthens education systems worldwide and responds to contemporary global challenges through education. The UNESCO AI CFT has been designed to complement and align with UNESCO's ICT Competency Framework for Teachers (ICT CFT) since 2018, drafting a separate framework to address the challenges and opportunities presented by AI systems. The aim of this framework is to guide teachers, educational communities and global teacher education systems in harnessing

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the educational benefits of AI. It encourages the cultivation of essential agency, knowledge, skills, attitudes and values necessary for addressing the risks and challenges linked with AI. The framework supports the responsible, ethical, fair and inclusive application of AI in education. The UNESCO AI CFT recognises the significant change in the knowledge, competencies, mindsets and principles that educators require to interact effectively and ethically with evolving AI tools, technologies and systems in the field of education. The draft document consists of 15 competencies with three broadly defined levels: *Acquisition*, *Deepening* and *Creation* and five competence areas. *Human-Centered Mindset; Ethics of AI; AI Foundation & Applications; AI Pedagogy; AI for Professional Development*. (https://www.unesco.org/sites/default/files/medias/fichiers/2023/12/UNESCO-Draft-AI-competency-frameworks-for-teachers-and-school-students_0.pdf).

Developing AI competence frameworks based on competence areas provides a flexible approach. New areas, such as human-centred mindset or maker-oriented competencies, can be incorporated, although clear explanations of terms and communication are crucial to ensuring understanding of the competence area. The domains in the framework may also overlap or possess fuzzy boundaries, making it imperative to establish precise definitions and delineations where needed. This type of framework allows for the adaptability and responsiveness of evolving AI in education.

3.3 | Process-driven (process competency model) frameworks

The next group of frameworks are process-driven or process competency models, which represent processes and corresponding roles in a job. The representation of these models may have a complex design in relation to AIED. To examine these frameworks, one could ask: What processes characterise AI application in the classroom? How can these processes be translated into competencies? Or how can AI competencies attend to these processes?

Two conceptual frameworks on human-AI hybrid approaches related to teachers' AI competencies are discussed in this section. The significance of conceptual frameworks is underscored by Antonenko (2015)), who views them as practical tools for structuring inquiries. Conceptual frameworks are defined as a synthesis of beliefs, assumptions, theories and concepts that provide guidance throughout the inquiry process. Unlike relying on existing theories or readily available models, a conceptual framework is meticulously constructed by the researcher. Serving as a tool for integrating practical contexts, theoretical aspects and inquiry methods, it plays a significant role in shaping the focus of inquiries or programmes. Importantly, conceptual frameworks are not rigid formulas but rather offer guidance for curriculum and program design tailored to specific contexts (Falloon, 2020).

The field of AIED introduces new responsibilities for both learners and teachers, necessitating teachers' attention, decision-making, action and observation. Holstein et al. (2020) explored a conceptual framework for the process of human-AI hybrid adaptivity in education. They investigated a set of essential components, including goals/targets, perception, action, decision policies and granularity/timing. These components examine how humans and AI systems can complement each other in distinctive ways:

1. **Goal Augmentation**: In this aspect, both the teacher and AI system collaborate to enhance their respective objectives, striving to achieve superior outcomes
2. **Perceptual Augmentation**: This component focuses on enhancing perceptual abilities. Both the human teacher and AI system work to improve their capacity to sense and understand information
3. **Action Augmentation**: Here, both the teacher and AI system strengthen their capabilities to take action or execute tasks more effectively
4. **Decision Augmentation**: This part concentrates on refining the decision-making process. The teacher and AI system collaborate to make better decisions by sharing their insights and expertise.
5. **Granularity and Timing**: The frequency or granularity at which the perception-action cycle is performed varies, ranging from once per task/step to once per conversation turn/ design iteration.

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Within each category, the teacher and AI systems enhance each other's abilities in real-time (Holstein et al., 2020). This framework offers a structured approach to understanding and facilitating collaboration between human and AI elements in the classroom (Table 4).

Molenaar (2022) further develops the automation model of teachers' adaptivity. Inspired by similar models used in the automotive and medical fields, this conceptual framework outlines the progression of control among the teacher, learner and technology across six distinct levels of automation. Teachers play a pivotal role in determining what specific organisational tasks they hand over the control to AI, fostering reciprocal relationships and facilitating communication between humans and artificial intelligence. This model emphasises the significance of two-way interaction between AI and teachers, as teachers possess valuable insights and knowledge to decide what tasks to hand over to AI technology and what tasks they consider beyond the reach of AI, such as providing elaborate feedback or additional instruction. This interaction is fundamental to the augmentation perspective on AI in education. This suggests that teachers require a platform to share their professional knowledge with AI, contributing to the vision of enhancing learning and teaching through hybrid human-AI systems (NOLAI Referentiekader, 2022). Based on the model, we developed a set of AI competencies for educators (see Table 5).

TABLE 4 Process-driven competencies adapted from Holstein et al. (2020) AI-Human hybrid adaptivity framework.

The process goes both ways and teachers should have competencies to enable reciprocal actions with AI system		
AI →	↔	← Educator
	1. Inform Instructional goals; 2. Perceive relevant information; 3. Distribute attention; 4. Interpret the information; 5. Perform instructional action; 6. Expand the range of actions available; 7. Enhance scalability and capacity for action; 8. Make more effective pedagogical decisions; 9. Adapt instruction (per task/step/turn/design iteration – prior/in the midst/afterwards of the activity).	

TABLE 5 Teachers' AI competencies adapted from Molenaar (2022) conceptual model 'The six levels of automation'.

Teacher knows:
- When/Why/How/to have full/partial control over the task.
- When/How/Why/Where/What to monitor.
- When/Why/How to cede/reclaim control over technology.
- When/Why/How technology provides supportive information.
- When/why/how technology controls specific tasks.
- Which tasks are controlled by technology.

Teacher Recognises:
- Technology signals.
- Supportive information.

Teacher is able to:
- Respond appropriately to technology signals.
- Have full/partial control over the task.
- Cede/reclaim control over technology.
- Interpret supportive information and respond appropriately.
- Monitor the technology and the process.

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An AI competence framework can be developed around fundamental processes or behavioural competencies that describe the adaptivity or relationship between AI, teachers and learners. These process competency frameworks, often utilised in industries such as automotive processes, can be adapted for educational use with some modifications. AI CFTs based on the process-driven/behavioural competencies developed by conceptual frameworks can have intricate designs. They demonstrate the dynamic relationships between AI, teachers and learners. Due to the difference between the nature of processes in the industrial and educational fields, the implementation of such competency models in the educational context should be careful when inspired by automation models while accommodating the unique processes within the field of education.

3.4 | AI-systems (specific job/work model) frameworks

Another approach to organise essential AI competencies for teachers involves grouping them around a job/work model (AI systems). Recent research has explored machine working frameworks that suggest categorising skills based on the specific AI models in use. These frameworks typically comprise three key components: machine operational competencies, user competencies and utilisation competencies. Correspondingly, the questions to analyse such frameworks are the following: What knowledge is necessary for the AI system operation? How this knowledge be applied in practice? Who are the users of this system? How should this system operation adhere to regulations of data safety, privacy and protection?

For instance, Carolus et al. (2023) conducted a study aimed at defining competencies for effective interactions with voice-based AI systems, addressing a notable gap in AI-related literacy models. Developers, designers and similar profession experts who self-reportedly developed or designed language-based AI systems in their work and have gained relevant experience in this area were interviewed. The answers resulted in the creation of a comprehensive Digital Interaction Literacy (DIL) competency model, which encompasses both universal and specific skills tailored to different user groups, including parents, teachers and developers. The DIL Model places a strong emphasis on enhancing user proficiency by fostering a foundational understanding of AI systems, including their operations and natural language processing capabilities. This, in turn, empowers users to make well-informed decisions and optimise their interactions with these systems. The primary objective of the DIL Model is to encourage conscientious engagement with AI systems by promoting awareness of both their benefits and challenges. This model equips practitioners with a guide for considering the perspectives of diverse users, allowing them to align their expectations with reality, extract more significant benefits from their interactions with AI systems and adapt their behaviour, such as by providing instructions for formulating appropriate requests (Table 6).

AI competence frameworks tailored to specific AI systems can be particularly beneficial for educators at the early stages of AI introduction to schools and for those who are seeking to master narrow AI competencies. The introduction of AI into education is a gradual process that depends on country and school policy, and such frameworks can be highly effective when educators work with specific AI models (e.g., application of ChatGPT in Su & Yang, 2023) introduced in their schools. Such AI competence frameworks can complement

TABLE 6 Digital Interaction Literacy (DIL) competency model adapted from Carolus et al. (2023).

Understanding functional principles	User group dependent competencies	Mindful usage
General functional knowledge Device and system handling Algorithm awareness AI-learning	Development Communication and teaching	Privacy literacy Persuasion literacy Device and system handling Reflection

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the broader Digital Competence Framework for teachers and will likely require a more specialised focus on AI proficiency.

3.5 | Competence level-driven frameworks

As AI continues to gain prominence in educational settings, the development of new AI competence frameworks becomes an evolving effort. To meet the emerging needs of AI integration in schools, an innovative competence-level-driven approach is being adopted. This approach is designed to seamlessly weave together various levels of competencies. It forges a comprehensive link between AI models, tools, applications, implications and attitudes, thus creating a multidimensional framework for AI competence and answering the following questions: What levels does this model consist of? Are there sub-competencies? Are AI competencies divided into components?

Sattelmaier and Pawlowski (2023) have introduced a Generative AI (GAI) competence framework tailored for K-12 schools, with a primary focus on equipping both teachers and students. This framework unfolds across multiple tiers, beginning with general, digital and domain-specific competencies, which include fundamental skills like critical thinking and data literacy. It progresses into more advanced AI competencies, comprising algorithms, programming, data literacy, AI techniques, AI technologies, AI development, application of AI, ethics of AI and social implications. Moreover, this framework introduces emerging competencies focusing on the art of effectively prompting large language models. This involves an understanding of different GAI model types, responsible utilisation of GAI tools and applications for content generation, thorough exploration of GAI implications, including risk assessment and problem-solving, and the development of AI attitudes, linked to the broader implications of AI (Figure 1).

Another example of competence level-driven framework is Ng et al. (2023) framework. The framework includes: technical and non-technical skills, and meaningful strategies to enhance students' AI competencies. The model suggests that proficiency in AI should extend beyond technical aspects (e.g., attitudes, skills and knowledge). It advocates for a comprehensive understanding that acknowledges a non-technical, critical competence that young people should acquire to ethically, safely and wisely navigate AI technologies. Taking a broader perspective, teachers are encouraged not to perceive AI competence as solely a distinct technological area but as an opportunity to cultivate other non-technical skills, including life and career skills, multidisciplinary skills and learning and innovation skills. Additionally, it presents a revised model for teachers to implement meaningful strategies for enhancing their students' AI competencies through four key processes: (1) teachers' professional engagement; (2) instructional support; (3) content choices across disciplines; and (4) students' learning competencies (Ng et al., 2023).

The competence level-driven approach is a systematic endeavour, providing depth, clarity and hierarchy to the relationships and connections within the framework, which might pose questions about its implementation in practice. As authors claim, such models are not complete because they cannot include all evolving tools (Sattelmaier & Pawlowski, 2023). It portrays the construction of competencies as blocks, layers or levels, offering an illustration of the complex nature of teacher education in the age of AI. With such a profound representation of competencies, prioritisation of competencies and consultations with experts could be important.



FIGURE 1 Generative AI (GAI) competence framework adapted from Sattelmaier and Pawlowski (2023).

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4 | DISCUSSION

The objective of this paper was to explore AI Competence Frameworks for Teachers (AI CFTs) and classify them within Competence Construct Claims. We provided a structured classification of available AI CFTs to understand their foci and characteristics, including AI competencies and the rationale behind their grouping to promote teacher professional development in AI (Table 7).

In our critical review on AI CFTs, we distinguished the following types of frameworks: (1) Existing competence model-oriented; (2) Competence areas/domains-oriented; (3) Process-driven (process competency model); () AI systems-driven (specific work/job model) and (5) Competence level-driven (multidimensional). All the frameworks under discussion were at the initial stage of their development. Due to the diverse and complex nature of artificial intelligence, the models were inherently intricate and required thoughtful consideration. It is remarkable that the frameworks that were devised from empirical data – bottom-up approach (Carolus et al., 2023; Kim & Kwon, 2023) – are more detailed and include descriptions of competencies at the micro level, and the frameworks produced from conceptual models are more general with broader areas (Higuera, 2019; Lorenz & Romeike, 2023). Nevertheless, all the frameworks need validation and real-life application, including investigation and the collection of evidence (Table 7).

Our classification of AI Competence Frameworks (CFTs) can be instrumental for researchers and policymakers in comprehending approaches to teacher competence development. Defining teachers' AI competence necessitates a comprehensive mapping, considering both top-down and bottom-up perspectives.

While policy-makers, researchers and app designers may contribute to a top-down approach in delineating teachers' AI competencies, the empirical and design-based nature of AIED requires a solid theoretical foundation. The adoption of theoretical frameworks serves as a unifying force, fostering shared concepts and terminology among researchers and designers, facilitating more efficient communication and innovation (Graziani et al., 2023). Holstein et al. (2020) emphasise the significant influence of theory in guiding the formulation of hypotheses, predictions, comprehension and design strategies within AIED. However, a notable disparity arises, as most AI competencies are derived from instructional theories or literature reviews rather than teachers' actual experiences. Consequently, teachers often perceive these proposed competencies as irrelevant to enhancing their teaching practice, leading to diminished motivation (Kim & Kwon, 2023).

On the other hand, it is essential that, through empirical studies, AI teachers' competencies are defined (bottom-up): what components/elements they comprise, which cognitive, action-related or motivational attributes they embrace. With the wide range of AI-tools, it is important to understand what kind of problems AI teachers' competencies will solve and what kind of solutions AI teachers' competencies convey. Another important question is how these competencies are developed, assessed or measured (Klieme et al., 2003). Further research is vital to assess the practical application of AI tools and techniques. Without first-hand experience of these tools and systems in action, it is challenging to predict the precise processes and skills needed for their smooth operation. By blending top-down expertise with bottom-up input, AI competency models can become more dynamic, relevant and adaptable to the evolving realities of the field.

As the field of AIED continues to evolve, several concerns should be addressed regarding the development of AI CFTs. First, it is imperative to acknowledge that AI extends beyond the realm of traditional digital technology. Educators' competencies should be well-coordinated to prevent redundancy and minimise potential confusion if AI competencies are infused in Digital Competence Frameworks for Educators. Additionally, the Digital Competence Frameworks establish specific target levels that educators are expected to attain. This raises a pertinent question: should AI CFTs also encompass prerequisites, mastery levels and a successive progression of proficiency or competency benchmarks to ensure educators' proficiency aligns with the demands of AI-enhanced education? Furthermore, understanding how educators are assessed for their AI competencies and how they can assess their students' AI literacy is crucial.

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TABLE 7 Classification scheme of AI CFTs (summary).

N	Construct claims of AI CFTs	Questions for the framework	Focus	Characteristics and strengths	Papers
1	Integrating AI competencies in existing teacher competence frameworks or theoretical models	• How can new AI competencies be included in the earlier frameworks? • What kind of links can be identified between existing competencies and the new ones?	Integration into already known theoretical models, general reference frameworks Technological Pedagogical Content Knowledge (TPACK) Mishra and Koehler (2006), Punie (2017), ICT CFT (UNESCO, 2018)	The frameworks are: • Built on the existing frameworks which are validated and structured; • Connecting AI competencies to existing domains; • Likely to have some rigidity, that can limit integration of innovative elements.	Higuera (2019), Kim and Kwon (2023), Lorenz and Romeike (2023), Ethical Guidelines on the Use of AI, 2022
2	Modelling new AI knowledge areas/ domains (Klieme et al., 2003)	• What are new knowledge areas? • What are essential new skills or attitudes that educators need?	New themes/domains, knowledge/skills/attitudes	The Frameworks are: • Integrating new specific areas related to AIED; • Flexible; • Defining emergent areas; • Likely to be easily modified.	Olari and Romeike (2021), UNESCO (2023)
3	Process-driven (Process competency model) (Rothwell & Lindholm, 1999)	• What processes characterize AI application in the classroom? • How can these processes be translated into competencies? Or how can AI competencies attend to these processes?	Relationships between actors, Augmentation (Conceptual frameworks on Human-AI hybrid approach)	The frameworks: • Represent automation process parsing; • Can represent complex processes; • Have intricate design; • Should be implemented carefully in education, due to the distinctive features of the educational process	Holstein et al. (2020), Molenaar (2022)

(Continues)

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TABLE 7 (Continued)

N	Construct claims of AI CFTs	Questions for the framework	Focus	Characteristics and strengths	Papers
4	AI systems-driven (Prifti et al., 2017)	• What knowledge is necessary for the system operation? • How this knowledge should be applied in practice? • Who are the users of this system? • How should this system operation adhere regulations of data safety, privacy and protection?	User manual approach: machine operational competencies, user competencies and utilisation competencies	The frameworks: • Present AI systems operation principles and users' involvement; • Help master narrow AI system; • Are applicable in countries and schools with gradual AI integration in education; • Can be complementary to other general reference frameworks; • Can provide AI proficiency in certain AI systems; • Can hinder broader understanding of AI and deter the use of other AI systems.	Carolus et al. (2023), Su and Yang (2023)
5	Competence level-driven (Klieme et al., 2003)	• What levels does this model consist of? • Are there sub-competencies? • Are AI competences divided into components?	Holistic, systematic approach	The frameworks: • Are multidimensional for more than one user's groups • Have profound representations of the competences; • Can be complicated to implement • Can be incomplete	Sattelmaier and Pawlowski (2023), Ng et al. (2023)

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Second, the intersection of AI and subject didactics needs a thoughtful examination. Presently, AI competencies are often framed as cross-curricular, transcending specific subjects. However, an evolving discourse proposes the inclusion of subject-specific AI competencies. This prompts a reflection on the necessity for these competency models to be underpinned by the theoretical framework of subject didactics. Educators must possess deep subject knowledge in addition to digital and AI competencies to effectively guide their students in domain specific learning. Knowledge on the integration of AI in teaching and learning is generally limited within the industry (Havinga & Clary, 2023). In a curriculum landscape often dominated by large publishers and app designers, engagement of educators in curriculum development is therefore vital. The right balance between subject expertise and AI proficiency is a challenge that must be addressed in AI CFTs.

Third, AI CFTs should include a specific implementation model which will ensure the stages of AI adoption, adaptation, appropriation, transformation, mastery and optimisation depending on the specific technology, context and educational objectives. Furthermore, resources required to implement AI competence frameworks in educational institutions is essential. This includes considerations of funding, infrastructure, teacher training and ongoing support.

Finally, it is essential to recognise that there is no perfect AI CFT. As these frameworks evolve, their effectiveness depends on the extent of AI knowledge considered necessary for teachers to acquire for the application of AI in education. The ongoing quest for the ideal AI competence framework should be driven by the vision of educators' AI proficiency. As AI competence areas continue to evolve, it is imperative to facilitate interdisciplinary discussions involving experts and subject teachers. AI CFTs should be improved and implemented to ensure that educators are equipped with a versatile combination of pedagogy, subject expertise, digital literacy and AI proficiency, ultimately empowering them to provide technologically enriched education for their students.

5 | CONCLUSION

This paper explored AI Competence Frameworks for Teachers (AI CFTs) and classified them within the Competence Construct Claims. While this classification into five distinct types reveals the diverse nature of existing frameworks, it also underscores the early stage of AI CFT development and the complexity of addressing the multifaceted aspects of AI. Future research is recommended to focus on the validation of frameworks, longitudinal analyses and assessments of AI CFTs to gauge the real-life impact, sustained effectiveness and global applicability of the frameworks. Additionally, investigating specific implementation models and including stages of AI adoption will contribute to refining and effectively implementing AI CFTs.

The review covers AI CFTs in scientific papers between 2019 and 2023 and may not reflect the most recent developments or perspectives, especially in the rapidly changing AI field. We acknowledge that information and understanding of a topic can evolve over time and we may unintentionally bring our own biases or preconceived notions to the review process, influencing the interpretation of the material.

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CONFLICT OF INTEREST STATEMENT

None declared.

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DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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